whether you chase a light beam, or run from it, or just stand still, the light retains its one fixed and never changing speed of 670 million miles per hour. But, we naturally ask, how can light possibly behave in such a strange manner?

Think about speed for a moment. Speed is measured by how far something goes divided by how long it takes to get there. It is a measure of space (the distance traveled) divided by a measure of time (the duration of the journey). Ever since Newton, space had been thought of as absolute, as being out there, as existing "without reference to anything external." Measurements of space and spatial separations must therefore also be absolute: regardless of who measures the distance between two things in space, if the measurements are done with adequate care, the answers will always agree. And although we have not yet discussed it directly, Newton declared the same to be true of time. His description of time in the *Principia* echoes the language he used for space: "Time exists in and of itself and flows equably without reference to anything external." In other words, according to Newton, there is a universal, absolute conception of time that applies everywhere and everywhen. In a Newtonian universe, regardless of who measures how much time it takes for something to happen, if the measurements are done accurately, the answers will always agree.

These assumptions about space and time comport with our daily experiences and for that reason are the basis of our commonsense conclusion that light should appear to travel more slowly if we run after it. To see this, imagine that Bart, who's just received a new nuclear-powered skateboard, decides to take on the ultimate challenge and race a beam of light. Although he is a bit disappointed to see that the skateboard's top speed is only 500 million miles per hour, he is determined to give it his best shot. His sister Lisa stands ready with a laser; she counts down from 11 (her hero Schopenhauer's favorite number) and when she reaches 0, Bart and the laser light streak off into the distance. What does Lisa see? Well, for every hour that passes, Lisa sees the light travel 670 million miles while Bart travels only 500 million miles, so Lisa rightly concludes that the light is speeding away from Bart at 170 million miles per hour. Now let's bring Newton into the story. His ideas dictate that Lisa's observations about space and time are absolute and universal in the sense that anyone else performing these